



Potential for mitigating global agricultural ammonia emission: A meta-analysis[☆]

Chaopu Ti ^{a, b}, Longlong Xia ^c, Scott X. Chang ^{d, b, **}, Xiaoyuan Yan ^{a, *}

^a State Key Laboratory of Soil and Sustainable Agriculture, Institute of Soil Science, Chinese Academy of Sciences, Nanjing, 210008, China

^b Department of Renewable Resources, University of Alberta, Edmonton, AB T6G 2E3, Canada

^c Institute for Meteorology and Climate Research (IMK-IFU), Karlsruhe Institute of Technology, Kreuzeckbahnstrasse 19, 82467, Garmisch-Partenkirchen, Germany

^d State Key Laboratory of Subtropical Silviculture, Zhejiang A & F University, Hangzhou, 311300, China

ARTICLE INFO

Article history:

Received 19 June 2018

Received in revised form

29 October 2018

Accepted 29 October 2018

Available online 5 November 2018

Keywords:

Ammonia emission

Agricultural system

Fertilizer source

Manure management

Reduction strategy

Meta-analysis

ABSTRACT

Ammonia (NH₃) emission from agricultural sources has contributed significantly to air pollution, soil acidification, water eutrophication, biodiversity loss, and declining human health. Although there are numerous strategies for reducing NH₃ emission from agricultural systems, the effectiveness of these measures is highly variable. Furthermore, the integrated assessment of measures to reduce NH₃ emission both from livestock production and cropping systems based on animal and crop type is lacking. Therefore, we conducted a global meta-analysis and integrated assessment of measures to reduce NH₃ emission from agricultural systems. Most of the studied mitigation strategies were effective in reducing NH₃ emission. In the livestock production system, dietary additive, urease inhibitor (UI), manure acidification and deep manure placement have the highest mitigation potential relative to other mitigation strategies, with reduction ranges of 35.1–54.2%, 24.3–68.7%, 88.8–95.0%, and 93.8–99.7%, respectively, relative to the control, while manure storage management could significantly reduce NH₃ emission by 70.0–82.1%. In the cropping system, fertilizer source, use of enhanced efficiency fertilizers, and method of field application are most effective for reducing NH₃ emission. The use of ammonium nitrate, controlled release fertilizer (CRF), and deep placement of fertilizers could reduce NH₃ emission by 88.3, 56.8, and 48.0%, respectively. Choosing a proper fertilizer is critical for decreasing NH₃ emission from cropping systems. We conclude that carefully planned and adopted strategies suited for local conditions are promising for minimizing NH₃ emission from agricultural systems on a global scale, while possible effects of those mitigation measures on the emission of greenhouse gases should be studied in the future.

© 2018 Elsevier Ltd. All rights reserved.

1. Introduction

Ammonia (NH₃) is a key component and plays an important role in the global nitrogen (N) cycle (Erisman et al., 2007; Galloway et al., 2003). About 40% of the current global food production would not exist without synthetic NH₃ production for the manufacture of N fertilizers (Pfromm, 2017; Reis et al., 2009). However, a large percentage of fertilizer N used for food production is lost to the environment in many different forms, including NH₃ emission to the atmosphere, owing to excessive N input to and inefficiencies

in the food production–processing–consumption chain (Galloway et al., 2008; Sutton et al., 2011). The NH₃ emitted to the atmosphere will cause air quality degradation and climate change, and damage human health. When the emitted NH₃ eventually returns to the Earth's surface through N deposition, it could result in soil acidification, water eutrophication, and biodiversity loss (Behera et al., 2013; Ellis et al., 2011; Krupa, 2003; Stevens et al., 2011). Additionally, when NH₃ is transformed into ammonium aerosols, the cost associated with the impact of fine particles (PM_{2.5}) on human health is up to US\$75 per kg of N from NH₃ pollution (Stokstad,

[☆] This paper has been recommended for acceptance by Joerg Rinklebe.

* Corresponding author.

** Corresponding author. Department of Renewable Resources, University of Alberta, Edmonton, AB T6G 2E3, Canada.

E-mail addresses: sxchang@ualberta.ca (S.X. Chang), yanxy@issas.ac.cn (X. Yan).

2014). The annual health cost caused by NH₃ emission in the European Union is estimated at US\$18–140 billion, mostly as a result of increased mortality associated with increasing atmospheric aerosol concentration (Brink and Grinsven, 2011).

Agricultural activities dominate the global NH₃ emission to the atmosphere (Bouwman et al., 1997; Paulot et al., 2014). The EDGAR v4.2 (Emissions Database for Global Atmospheric Research) emission inventory estimates that 81% of the global NH₃ emitted to the atmosphere comes from agriculture sources (Van Damme et al., 2014). Without significantly changing the N management practices in the current food production system, the increasing demand for food will result in greater NH₃ emission (Aneja et al., 2001; Erisman et al., 2008). Mitigating NH₃ emission, especially agricultural NH₃ emission is urgently needed to improve both environmental and human health. Mitigation strategies for NH₃ emission from both cropping and livestock systems have been extensively studied. For example, a recent meta-analysis evaluated NH₃ emission by strategies such as '4R Nutrient Stewardship', farming practices, and enhanced fertilizer use efficiency (Pan et al., 2016). They found that mitigation measures such as deep placement of fertilizers and irrigation significantly decrease NH₃ emission from cropping systems by 75 and 55%, respectively, relative to the control in their studies. However, their study did not consider the effects of strategies on individual crops, and the use of organic fertilizers such as manure was not included. The effect of strategies used on NH₃ emission reduction may differ for different crops. For example, the application of CRF has a stronger effect on reducing NH₃ emission in rice and maize than in wheat production systems, and the effect of deep placement of fertilizer on NH₃ emission also varies significantly with crop type (Xia et al., 2017a). Moreover, systematic approaches for mitigating NH₃ emission from vegetable and pasture systems are lacking, while a large amount of NH₃ is emitted from vegetable production systems because of heavy use of fertilizer N (Ti et al., 2015).

Livestock production systems are another main agricultural source of NH₃ emission. Numerous measures have been developed to mitigate these emissions, most of these strategies are focused on one specific strategy or animal production system (Chai et al., 2016; Mendes et al., 2017; Nakhshiniev et al., 2014; Wang et al., 2017). In recent years, several systematic analyses have determined the effect of mitigation measures on NH₃ emission from livestock production systems (Bittman et al., 2017; Hou et al., 2015; Loyon et al., 2016; Xu et al., 2017). For example, Hou et al. (2015) conducted a meta-analysis and an integrated assessment of the effect of mitigation measures on NH₃ emission from the whole manure management chain. They found that proper farm scale combinations of mitigation measures are important for minimizing the impact of

livestock production on global NH₃ emission. However, comprehensive analyses of the effect of different mitigation strategies by animal type in the livestock production process are necessary. Furthermore, there has been no comprehensive analysis of the effect of management techniques on NH₃ emission from both livestock and crop production systems.

Effective measures are urgently needed to guide stakeholders and policymakers on minimizing NH₃ emission from agriculture. Thus, to understand the potential for mitigating agricultural NH₃ emission on a global scale, and to fill the current knowledge gaps described above, we conducted this study 1) to evaluate the effect of mitigation options on NH₃ emission from agricultural systems including cropping and livestock production systems, based on individual crop/animal types, through a meta-analysis, and 2) to offer recommendations for future research based on this meta-analysis and literature review.

2. Materials and methods

2.1. Selection of mitigation measures to include in the meta-analysis

Mitigation measures for NH₃ emission from agricultural systems are wide-ranging. Typical cropping and fertilizer management practices associated with NH₃ emissions include the use of CRF, nitrification inhibitors (NIs), urease inhibitors (UIs), split application and deep placement of fertilizers, and irrigation, among others. In this meta-analysis, all of the above cropland management practices were incorporated into four separate datasets based on peer-reviewed publications (Table 1). Crops such as rice (*Oryza sativa* L.), maize (*Zea mays* L.), wheat (*Triticum aestivum* L.), vegetables, fruits, and pasture were selected as crop production systems for study.

Ammonia emissions from livestock production systems mainly come from animal excreta in housing, manure storage, and manure applied to land. Animal feed composition and feed management also has a strong influence on animal performance and the chemical composition of dung and urine, and thus on the emission of NH₃. Therefore, based on the recent NH₃ emission guidance from the UNECE (United Nations Economic Commission for Europe) Task Force on Reactive Nitrogen (Bittman et al., 2014), our study will be focused on NH₃ emission from livestock production systems, including livestock feeding strategies, animal housing techniques, manure storage techniques, and manure application techniques (Table 1, and Table S1). Production systems for livestock such as pig, cattle, and poultry were considered in this meta-analysis because sufficient data could be obtained and NH₃ emission from these

Table 1
Selected mitigation measures for reducing NH₃ emission from agricultural sources (further details of each measure are shown in Tables S1 and S2).

System	Strategies	Details of measure
Livestock production	Feeding strategies	Dietary acidifier, dietary bacteria (clostridium butyricum, bacillus licheniformis, bacillus subtilis, bacillus amyloliquefaciens, enterococcus faecium), yucca extract and dietary additives (cinnamon extracts, Ferix-3, Al ⁺ Clear, zeolite, copper silicate nanoparticles, tannin, Zn, maize distillers grains with soluble, gypsum), and crude protein content reduction
	Housing strategies	Use of adsorbent, mechanical management (air and manure cleaning systems), floor management, and use of additives and urease inhibitors
	Storage strategies	Cover of stored manure, use of acidifiers and additives, turning of manure piles, reduced crude protein content, aeration in manure piles, and compaction of manure
	Field management	Irrigation, anaerobic digestate of manure, and separation of solid and liquid manure reduced crude protein content, deep placement, incorporation, injection, band spreading
Cropland	Fertilizer source	Ammonium nitrate, ammonium sulphate, diammonium phosphate, calcium ammonium nitrate, ammonium bicarbonate, monoammonium phosphate, urea ammonium nitrate, urea phosphate, manure 50%, manure 75%, manure 100%, residue retention, and urea
	Enhanced efficiency fertilizers	Controlled release fertilizer, fertilizer with nitrification inhibitors and urease inhibitors
	Field management	Split application and deep placement of fertilizers, and irrigation

animals dominated total animal NH₃ emission (Bouwman et al., 1997).

2.2. Literature search and database compilation

Peer-reviewed publications from January 1980 to September 2017 that are related to NH₃ emission mitigation measures such as fertilizer application and manure management were extracted from the ISI Web of Knowledge (www.isiwebknowledge.com) database, China National Knowledge Infrastructure (CNKI) database, and Google Scholar by basic and advanced searches. Terms including NH₃ emission management, NH₃ mitigation, fertilizer management, fertilizer NH₃ emission, fertilizer volatilization, fertilizer N source, deep placement, the use of UI, the use of NI, NH₃ emission from cropland, irrigation impacts on NH₃ emission, and CRF application in titles, abstracts and keywords in articles were used to search for publications on cropland management practices. Terms including manure field application, dietary additive, crude protein content reduction, manure cover, manure acidification, manure turning, manure aeration, NH₃ emission from animal, and compaction were used to search for publications on livestock production system management practices.

Original data in publications were obtained from tables and figures. Data in figures were obtained by digitizing the figures using the GetData Graph Digitizer 2.22 (<http://getdata-graph-digitizer.com>). Database compilation followed the following criteria: data from field-, pilot-, and laboratory-scale studies were included, and studies have control treatments (such as no fertilizer application, fertilizer types compared with urea, manure storage without cover, and feed without additives) to allow assessment of treatment effects, with mean values of NH₃ emission for each treatment available. Other data such as manure pH, and animal weight were also collected from the publications.

Based on the database compilation criteria described above, a total of 1091 observations (587 observations for livestock systems and 504 observations for cropping systems) from 268 studies (96 studies for livestock systems and 172 studies for cropping systems) were collected from countries such as Australia, Belgium, Canada, China, Denmark, France, Germany, Hungary, Spain, the Netherlands, UK and USA to analyze the effect of mitigation measures on NH₃ emission from agricultural sources (see Tables S1–S2). The literature used in this study is presented in the supplementary file.

2.3. Data analysis

The effect of individual mitigation measures on agricultural NH₃ emission was calculated by comparing the result of control and treatment groups using the following equation:

$$\ln R = \ln \frac{X_t}{X_c}$$

where $\ln R$ is the effect size calculated as the natural log of the response ratio for each comparison, X_t is the mean value of NH₃ emission in the treatment group with a mitigation measure, X_c is the mean value of NH₃ emission in the control group without the mitigation measure. Mitigation efficiency was reported as the percentage change $(R - 1) \times 100$ of NH₃ emission under a treatment. Thus, a negative or positive percentage change indicates that the studied management practice can reduce or increase NH₃ emission, respectively. The effect sizes were weighted by the method described in Xia et al. (2017a). The meta-analysis of the mean effect size and the 95% confidence intervals (CIs) of each effect were calculated by MetaWin version 2.1 (Rosenberg et al., 2000). The mitigation efficiency of NH₃ emission was considered

to be significantly positive or negative if the 95% CIs did not overlap with zero, while the effects for which the 95% CIs overlap with zero were not significant.

Differences in other parameters such as the mean pH of manure in the treatment and control groups were analyzed using a one-way analysis of variance (ANOVA), followed by the least significant difference (LSD) test. In this study, $P < 0.05$ was considered statistically significant. The above statistical analyses were conducted using SPSS 17.0 (SPSS Science, Chicago, IL, USA).

3. Results

3.1. Effects of mitigation measures on NH₃ emission from cropping systems

The application of the following chemical fertilizers: ammonium nitrate, ammonium sulphate, urea phosphate, calcium ammonium nitrate, monoammonium phosphate, and diammonium phosphate could significantly reduce NH₃ emission by 88.3, 82.9, 76.2, 67.3, 51.9, and 51.2%, respectively, compared with the use of urea (Fig. 1). When comparing with 50, 75 and 100% substitution of livestock manure for synthetic N fertilizer (with an equivalent total N rate), only the 100% substitution significantly reduced NH₃ emission (Fig. 1). The retention of crop residues on the soil surface is a common feature in conservation farming; however, as one of the important sources of fertilizer N, residue retention such as wheat straw and leaf vegetable, could increase NH₃ emission by 22.3% on average, but not significant as compared with the control. The average reduction of NH₃ emission by all fertilizer sources was the highest for maize (81.2%), followed by pasture (79.9%), vegetables (68.3%), and wheat (41.7%).

On average, the application of CRF and UI significantly reduced NH₃ emission by 56.8 and 53.1% (Fig. 2). In contrast, NI application significantly increased NH₃ emission by 42.6%. The reduction in NH₃ emission by the application of CRF was greater in pasture and rice than in vegetable and maize production systems. The effect of UI on reducing NH₃ emission also varied significantly with crop type, with the highest for rice but the lowest for maize. Overall, the application of NI increased NH₃ emission.

Most selected fertilizer application managements could decrease NH₃ emission (Fig. 3). Deep placement and irrigation significantly reduced NH₃ emission by 48.0 and 38.8%. However, management of split application fertilizers has no significant reduction effect. Among crop types, significant reductions through fertilizer management were achieved in wheat and maize.

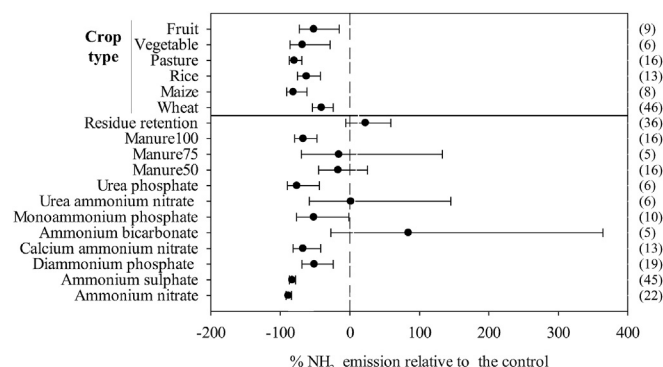


Fig. 1. Effects of fertilizer source on NH₃ emission from cropping and on production systems of major crops. The comparison is made relative to the control (urea as the control). Points show means of treatments, bars show the 95% confidence intervals (CIs). Values in parentheses indicate the number of observations.

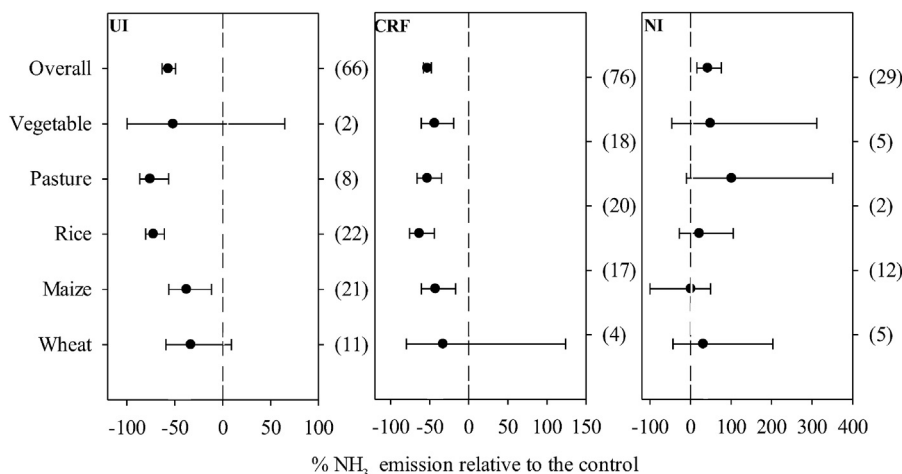


Fig. 2. Effects of urease inhibitors, controlled release fertilizer, and nitrification inhibitors on NH_3 emission in production systems of major crops. The comparison is made relative to the control. Points show means of treatments, bars show the 95% confidence intervals (CIs). Values in parentheses indicate the number of observations.

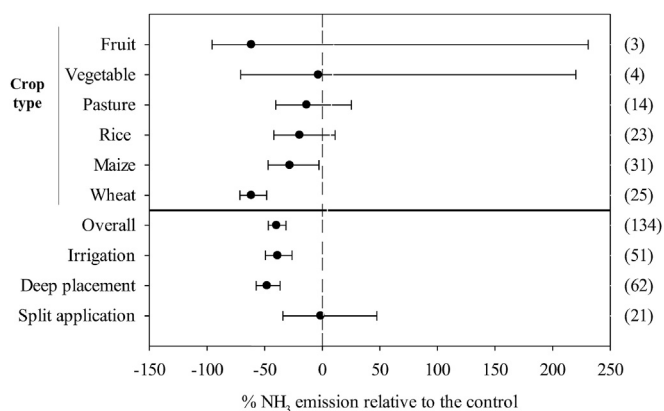


Fig. 3. Effects of field application management on NH_3 emission from cropping and in production systems of major crops. The comparison is made relative to the control. Points show means of treatments, bars show the 95% confidence intervals (CIs). Values in parentheses indicate the number of observations.

3.2. Effects of mitigation measures on NH_3 emission from livestock production systems

Effects of mitigation practices on NH_3 emission varied of livestock production. The average reduction of NH_3 emission by mitigation measures used by feeding strategies was 39.6% (Fig. 4a). The application of dietary additives (details of the application of dietary additives are shown in Table S1) was the most effective option for reducing NH_3 emission (45.5%), followed by crude protein content reduction (42.8%), while the use of dietary acidifier has no significant effect on NH_3 emission (8.6%). The reduction in NH_3 emission by mitigation measures through feeding strategy was greater in pig (42.2%) than in poultry (39.7%), while reduction effect on cattle was not significant.

For animal housing strategies, the application of adsorbent, additives, UI, and floor and mechanical management reduced NH_3 emission, by an average of 46.0% (Fig. 4b). The use of mechanical management is seen as the most effective mitigation measure to reduce NH_3 emissions (73.6%). Housing management decreased NH_3 emission for individual animals. The application of mitigation measures by housing strategies could significantly reduce NH_3 emission by 32.2%, 48.9% and 53.5% for cattle, poultry, and pig respectively.

Techniques used by manure storage showed a significant reduction potential of 62.2%, with a 95% CI ranging from 58.5 to 65.6% (Fig. 4c). Side-by-side comparisons showed that techniques such as covering the manure, the application of acidifiers and additives, could significantly reduce NH_3 emission by 92.5 and 29.3%. Manure aeration and turning increased NH_3 emission by manure storage strategies (2.3 and 108%, respectively), while not significant different compared with the control. Among selected animals, greater reductions through abatement measures were achieved in cattle and poultry than in pig (Fig. 4c).

Mitigation measures such as deep placement, incorporation, injection, and irrigation could significantly reduce NH_3 emissions by 98.7, 53.2, 67.5, and 73.6%, respectively, compared with the control. Overall, the average reduction of land application strategies was 60.7%, with 95% CIs ranging from 53.5 to 66.6% (Fig. 4d). The effect of land application strategies on reducing NH_3 emission also varied significantly with animal type, with the highest for poultry (74.3%) and the lowest for cattle (58.9%).

3.3. Summary of strategies with a high mitigation potential for agricultural NH_3 emission

There are large potentials to mitigate NH_3 emission using various management practices. Strategies such as the use of ammonium nitrate, ammonium sulphate, urea phosphate, calcium ammonium nitrate, 100% substitution of manure N for chemical fertilizer N, CRFs, fertilizers with UI, mono-ammonium phosphate, and diammonium phosphate could reduce NH_3 emission more than 50% from cropland compared with the control. The most effective method for reducing NH_3 emission from livestock production systems is manure storage technique, including the use of acidifiers and cover. Overall, fertilizer type is much more critical than land application strategy for reducing NH_3 emission from cropland. In contrast, field land application strategies such as deep placement of manure, manure injection, and irrigation are more efficient than feeding and housing strategies for lowering NH_3 emission from livestock production systems.

4. Discussion

To meet the growing demand for food and biofuels, global crop production by 2050 will have to double from 2005 levels, which means that the global N consumption would need to increase by 2.7

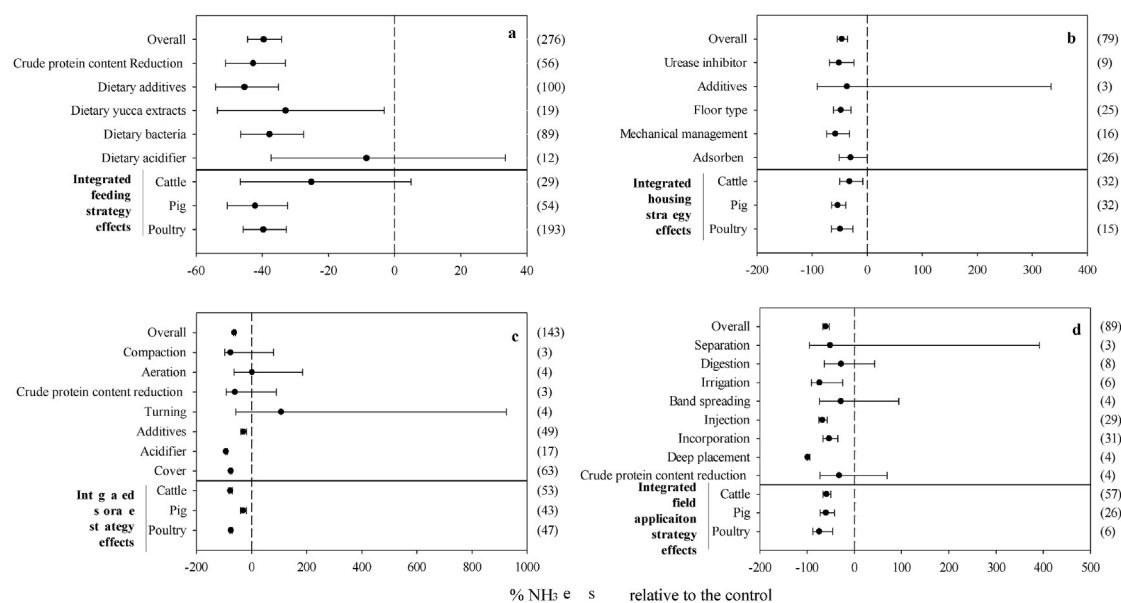


Fig. 4. Effects of strategies on NH₃ emission from livestock production systems and major animals. The comparison is made relative to the control. Points show means of treatments, bars show the 95% confidence intervals (CIs). Values in the parentheses indicate the number of observations a, feeding strategies; b, housing strategies; c, manure storage strategies; and d, manure field application strategies.

times to achieve the required food production (Tilman et al., 2011; Tilman et al., 2002). Therefore, increasing N input to agriculture systems will result in large quantities of N being lost in different forms, including in the form of NH₃ emission (Robertson et al., 2013). Increased NH₃ emission not only reduces the N available for crop production but also causes a range of environmental problems. Thus, systematic analysis of the effect of NH₃ emission mitigation measures on reducing NH₃ emission from agricultural systems is critically needed. This is the first comprehensive meta-analysis of literature data on the effect of management techniques on NH₃ emission from both cropping and livestock production systems.

4.1. Measures for reducing NH₃ emission from agriculture systems

A large number of management practices have been used for reducing NH₃ emission from agricultural systems. The efficiency of typical measures for specific animal/crop production systems has been extensively studied (Chai et al., 2016; Huang et al., 2016; Trabue and Kerr, 2014; Webb et al., 2014). This meta-analysis indicates that NH₃ emission from cropping systems could be significantly reduced by the application of fertilizers such as ammonium nitrate and ammonium sulphate compared to the application of urea fertilizers. Ammonia emission from the application of urea-based fertilizers is much greater than those from other fertilizer types selected in this study because the rapid hydrolysis of urea will cause a localized increase in pH, which enhances NH₃ emission (Bittman et al., 2014). The mitigation effect in reducing NH₃ emission from maize was much higher than that from rice production systems because paddy soils have a higher capacity to transform NH₃ to NH₄⁺ and adsorb NH₄⁺-N on to soil particle surfaces (Khalifa and Zidan, 2001). Additionally, manure is used as one of the fertilizer sources in cropping systems, and land application of manure could significantly reduce NH₃ emission. A large reduction (67.0%) in NH₃ emission can be achieved by 100% substitution (with an equivalent total N rate) of livestock manure for synthetic N fertilizer (urea) (Fig. 1), implying that appropriate livestock manure

application to farmland could reduce atmospheric N pollution. Substituting manure for chemical fertilizer may promote the microbial immobilization of bioavailable N, thus improving N uptake and crop yield. In addition, manure substitution may benefit the improvement of soil properties, such as soil structure, water holding capacity of soil and microbial activity, consequently enhancing crop yield with lower NH₃ emission (Xia et al., 2017b). Therefore, proper combinations of mitigation measures in both livestock and cropping systems are important for minimizing global emissions of NH₃ from agricultural systems. Moreover, the use of manure could improve crop productivity, and increase soil organic carbon storage (Xia et al., 2017b).

Other measures such as the use of enhanced efficiency fertilizers also affect NH₃ emission from cropland. For example, CRF and UI significantly decreased NH₃ emission from cropland in this study because of a better synchronization of crop N demand and supply and a decrease in urea hydrolysis were the mechanisms for the reduction in NH₃ emission (Huang et al., 2016; Sommer et al., 2004). The application of UIs in livestock production systems is effective to decrease NH₃ emission due to UI delaying the hydrolysis of urea and volatilization of NH₃ into the atmosphere (Varel et al., 1999). In contrast, application of NI increased NH₃ emission by 42.6%, with 95% CIs ranging from 15.6 to 76.0% from cropland (Fig. 3), in agreement with results from Qiao et al. (2015), Pan et al. (2016), and (Xia et al., 2017a). The NIs inhibit the conversion of NH₄⁺ to NO₃⁻, the retained NH₄⁺ in the soil is available for volatilization and can thus increase NH₃ emission (Chen et al., 2008; Di et al., 2007). The NI application can increase or decrease NH₃ emission depending on the soil properties and weather conditions, temperature in particular (Kim et al., 2012).

The effect of mitigation measures on NH₃ emission in livestock production systems has also been extensively studied (Hou et al., 2015; Wang et al., 2017). However, the effect of feeding strategies such as the use of dietary acidifiers, dietary bacteria, and dietary additives such as tannin, liquid alum, and maize distiller's dried grains with solubles (DDGS) on NH₃ emission has not been synthesized in previous studies. Our meta-analysis showed that the

largest mitigating potential for NH_3 emission reduction by feeding strategies is the use of dietary additives (45.5%), which is higher than the reduction effect by lowering dietary crude protein content (Fig. 1). Feed additives can change the microbiological environment in the animal digestive tract, the nutrient composition of feces, and the gaseous emission from the digestive tract and feces (Li et al., 2012; McCrory and Hobbs, 2001).

The emission of NH_3 occurs when manure and chemical fertilizers are applied in the field. Land management practices such as deep placement of fertilizers and irrigation can reduce NH_3 emission from cropland. A recent review indicated that deep placement of fertilizers such as liquid manure and NH_4^+ -based fertilizers could reduce NH_3 emission especially on alkaline or dry soils, and under high air temperature (Nkebiwe et al., 2016). Irrigation immediately following fertilizer application has been shown to reduce NH_3 emission by up to 70% because the irrigation water washes the N into the soil allowing NH_4^+ to be adsorbed on cation exchange sites (Oenema and Velthof, 1993) and thus reducing the amount of NH_4^+ available for NH_3 emission. On the other hand, split application of fertilizer N only reduced NH_3 emission by 1.4% from cropland. Although split application improves the synchronization between N supply and crop demand, the dense canopy in the more vigorous growth period may even act as a sink for NH_3 because of the NH_3 compensation point of plants (Bash et al., 2010; Farquhar et al., 1979; Zhang et al., 2010). The effects of fertilizer application practices on reducing NH_3 emission are much lower in vegetable than in wheat and fruit production systems; the lower reduction in NH_3 emission from vegetable fields is associated with intensive irrigation and split application of fertilizers (Ti et al., 2015).

4.2. Assessment of measures to reduce NH_3 emission from agricultural systems

Global NH_3 emission originating from livestock and cropland accounted for more than 50% of total NH_3 emission in 2008, with 13.3% from domestic animals and 43.3% from agricultural soils and crops (Sutton et al., 2013). Studies such as Wang et al. (2017), Hou et al. (2015) and Xu et al. (2017) were performed with a focus on mitigating NH_3 emission from livestock production systems. In addition, strategies such as irrigation, N application method and CRF, from cropland have been analyzed (Pan et al. (2016) Xia et al. (2017a)). However, global systematic synthesis studies on mitigation strategies for NH_3 emissions from the whole agricultural system are lacking. For example, N not incorporated into animal protein or into milk is excreted in dung and urine in livestock production systems, and then N in these excreta is lost as NH_3 when applied directly to cropland. In addition, the uptake into the crop could be consumed directly in livestock production systems as feed and manure additives could influence NH_3 emission.

Furthermore, assessment of measures to reduce NH_3 emission from individual animals or crops is sometimes inconclusive from previous studies. In fact, the effect of abatement measures on different animals has a large variability (Fig. 4). Meanwhile, a meta-analysis of the effects of feeding strategies with high potentials for NH_3 emission reduction such as dietary yucca extract, dietary acidifiers in livestock production systems were first reported in this study. Therefore, it has been essential to demonstrate new strategies to reduce NH_3 emissions (Reis et al., 2015).

4.3. Effects of mitigation strategies on agricultural products

To meet the increasing demand for agricultural products while protecting the environment at the same time, management practices that enhance N use efficiency and minimize N loss are urgently needed (Chen et al., 2011; Sheppard and Bittman, 2015). Use of CRF

and UI are important measures to reduce NH_3 emission from cropland. Xia et al. (2017a) indicated that those practices significantly increased grain yield and net economic benefit by 2.9–12.6% and 1.3–10.0%, respectively, compared to traditional N management. Furthermore, deep placement of fertilizers could be used as a tool to mitigate NH_3 emission and increase nutrient uptake (Nkebiwe et al., 2016).

In animal production systems, NH_3 emission can be reduced without losing production through the use of different NH_3 emission mitigation measures (Webb et al., 2014). Similarly, the mitigation measures used to reduce NH_3 emission did not impact on average daily gain of animal body weight by the housing and feeding strategies compared to the control treatment (Fig. 5). Caution must be taken to minimize the negative impact on animal performance while reducing NH_3 emission. According to Hou et al. (2015), a 0.5–2% reduction in dietary crude protein for diets with a mean crude protein content of 14.4% and a 4% reduction for diets with a mean crude protein content of 18.1% reduced NH_3 emission from the whole manure management chains.

This meta-analysis shows that measures used to reduce NH_3 emission also decreased the pH of gastrointestinal contents. The measures used to reduce NH_3 emission such as the addition of acidifiers by feeding strategies can improve the intestinal micro-ecological environment to reduce NH_3 emission (Jia et al., 2009). Additionally, the mitigation measures taken in this study decreased the pH of animal excreta (Fig. 6). The pH of the livestock manure controls the equilibrium between NH_3 and NH_4^+ in the liquid contained in the manure. Thus, there is a greater potential for NH_3 volatilization at higher pH (McCrory and Hobbs, 2001).

4.4. Limitations and implications for mitigating NH_3 emission

Our meta-analysis indicates that NH_3 emission from agriculture could be markedly reduced by most of the selected measures studied in this paper. However, NH_3 emissions are affected by many physical, chemical, and biological factors from croplands (Huang et al., 2016). Although options for NH_3 reduction at the various stages of livestock manure production system are interdependent, controlling emissions from application of manures to land is particularly important because these emissions are generally a large component of total livestock emissions and land application is the last stage of manure handling. Otherwise, reduction in N-excretion rates from livestock has the most direct effect on emissions. In addition, for manure management, physical characteristics

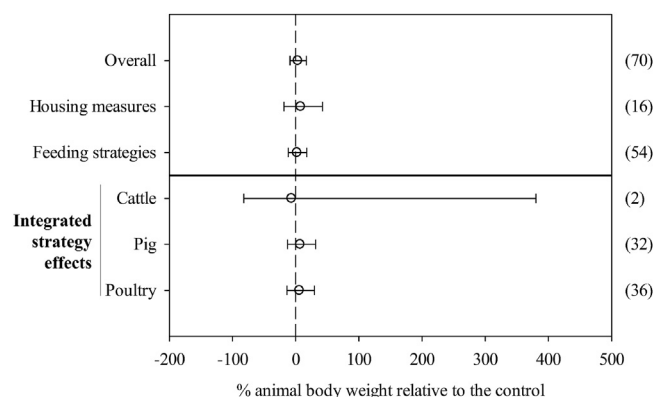


Fig. 5. Effects of housing and feeding management on average daily gain of animal body weight. The comparison is made relative to the control. Points show means of treatments, bars show the 95% confidence intervals (CIs). Values in parentheses indicate the number of observations.

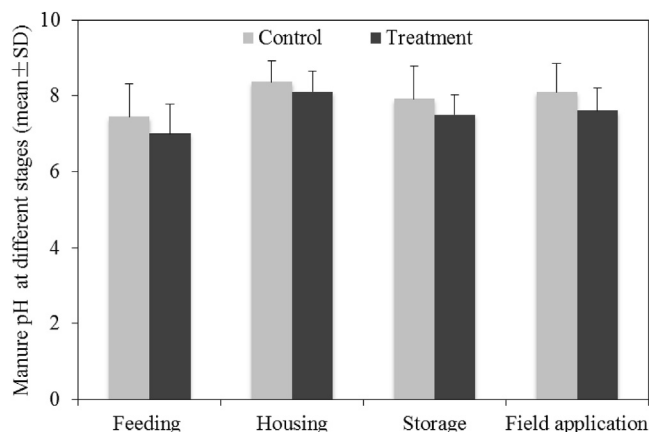


Fig. 6. Effects of integrated strategies on pH of manure in livestock production systems.

such as bedding materials are more important than chemical characteristics in determining NH_3 emissions from barn floors, and the removal efficiencies of NH_3 by acid scrubbers depending on their pH-set values (Gilhespy et al., 2009; Petersen et al., 2014). Therefore, the mean effect size and 95% CI of each selected measures in this study varied largely due to data obtained from different references. For example, the mean emission reduction of manure incorporation in fields was 53.2%, and the 95% CIs ranged from 35.1% to 66.3% (Fig. 4d). In addition, NH_3 abatement efforts of specific mitigation measures may differ due to implemented measures and policies. For instance, the applicability of measures depends strongly on farm structures, and not all measures can be implemented by small farms, which limits overall reduction potential (Wu et al., 2018). In addition to technical measures, emissions could also be reduced through policies that affect animal numbers (Stehfest et al., 2009).

In the future, the impact of a combination of mitigation measures on NH_3 emission should be studied. Several studies explored the impact of treatment technologies on NH_3 emission from livestock production systems through scenario analysis (Hou et al., 2017; Hou et al., 2015; Xu et al., 2017). For example, farmers applying manure to the field with methods such as injection or direct incorporation of manure into the soil is the most effective mitigation scenario for reducing NH_3 emission from manure generated by livestock production systems in China (Xu et al., 2017), consistent with the result from our meta-analysis. Furthermore, mitigating NH_3 emission needs to be carefully planned and adapted to local conditions because NH_3 emissions are dependent on environmental factors such as weather and soil conditions (Sutton et al., 2013).

Most of the mitigation measures were effective in reducing NH_3 emission, and measures such as the application of CRF and deep placement could significantly reduce N leaching and runoff (Xia et al., 2017a). However, attention should be paid to the possible effect of these mitigation measures on the emission of N_2O and CH_4 , which are potent greenhouse gases (Webb et al., 2004; Xia et al., 2017a; Xu et al., 2017). For example, nitrification–denitrification treatment decreased NH_3 emissions, but increased GHG emission (Hou et al., 2017). In addition, life-cycle analysis of fertilizer production and assessments of associated NH_3 emission is needed to develop effective mitigation options. Moreover, given the differences in mitigation efficiency among crop and livestock types, it may be more cost-effective to strategically reduce NH_3 emission based on the ecological region to meet the annual national emission reduction target.

5. Conclusions

Here, we first present a global meta-analysis and assessment of measures to reduce NH_3 emission from livestock production and cropping systems using 268 published results. Despite the aforementioned variability in the effect size of different mitigation measures, significant reduction efficiencies for NH_3 emission were observed for most of the mitigation strategies. In livestock production system, feeding management with dietary additives and reduction in crude protein content decreased NH_3 emission by 45.5 and 42.8%, respectively, compared with the control. NH_3 emission could be reduced by 62.2% by the use of cover, acidifiers and additives. In addition, livestock manure application through deep placement and injection could reduce NH_3 emission by 98.7 and 67.5%. In the cropland, application of chemical fertilizer type such as ammonium nitrate, ammonium sulphate and urea phosphate reduced NH_3 emission by 88.3, 82.9 and 76.2%, respectively, related to the control (the use of urea). Methods used for the application of chemical fertilizers such as deep placement and cropland management practices such as irrigation decreased NH_3 emission by 48.0 and 38.8%, respectively. The use of UIs and CRFs effectively decreased NH_3 emission but the use of NIs increased NH_3 emission by 42.6%. We conclude that there are great potentials for mitigating NH_3 emission from both livestock production and cropping systems. The economic feasibility for implementing the management practices analyzed above should be studied through a cost-benefit analysis in the future.

Conflicts of interest

We declare no competing interests.

Acknowledgements

This study was financially supported by the National Key Research and Development Program of China (No. 2018YFC0213304), the Key Research Program of the Chinese Academy of Sciences (No. ZDRW-ZS-2016-5-1), and the China Opportunity Fund at the University of Alberta (RES0031740). The first author also appreciates the Chinese Academy of Sciences for providing funds to support her study leave in Canada.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2018.10.124>.

References

- Aneja, V.P., Roelle, P.A., Murray, G.C., Southerland, J., Erismann, J.W., Fowler, D., Asman, W.A.H., Patni, N., 2001. Atmospheric nitrogen compounds II: emissions, transport, transformation, deposition and assessment. *Atmos. Environ.* 35, 1903–1911.
- Bash, J.O., Walker, J.T., Katul, G.G., Jones, M.R., Nemitz, E., Robarg, W.P., 2010. Estimation of in-canopy ammonia sources and sinks in a fertilized zea mays field. *Environ. Sci. Technol.* 44, 1683–1689.
- Behara, S.N., Sharma, M., Aneja, V.P., Balasubramanian, R., 2013. Ammonia in the atmosphere: a review on emission sources, atmospheric chemistry and deposition on terrestrial bodies. *Environ. Sci. Pollut. Control Ser.* 20, 8092–8131.
- Bittman, S., Dedina, M., Howard, C.M., Oenema, O., Sutton, M.A., 2014. Options for Ammonia Mitigation: Guidance from the UNECE Task Force on Reactive Nitrogen. Centre for Ecology and Hydrology, Edinburgh, UK.
- Bittman, S., Sheppard, S.C., Hunt, D., 2017. Potential for mitigating atmospheric ammonia in Canada. *Soil Use Manag.* 33, 263–275.
- Bouwman, A.F., Lee, D.S., Asman, W.A.H., Dentener, F.J., VanderHoek, K.W., Olivier, J.G.J., 1997. A global high-resolution emission inventory for ammonia. *Global Biogeochem. Cycles* 11, 561–587.
- Brink, C., Grinsven, H.v., 2011. The European Nitrogen Assessment: Sources, Effects and Policy Perspectives. Cambridge Univ. Press, Cambridge chap. 22.
- Chai, L.L., Krobek, R., MacDonald, D., Bittman, S., Beauchemin, K.A., Janzen, H.H.,

- McGinn, S.M., Vanderzaag, A., 2016. An ecoregion-specific ammonia emissions inventory of Ontario dairy farming: mitigation potential of diet and manure management practices. *Atmos. Environ.* 126, 1–14.
- Chen, D., Suter, H., Islam, A., Edis, R., Freney, J.R., Walker, C.N., 2008. Prospects of improving efficiency of fertiliser nitrogen in Australian agriculture: a review of enhanced efficiency fertilisers. *Aust. J. Soil Res.* 46, 289–301.
- Chen, X.P., Cui, Z.L., Vitousek, P.M., Cassman, K.G., Matson, P.A., Bai, J.S., Meng, Q.F., Hou, P., Yue, S.C., Romheld, V., Zhang, F.S., 2011. Integrated soil-crop system management for food security. *Proc. Natl. Acad. Sci. U. S. A.* 108, 6399–6404.
- Di, H.J., Cameron, K.C., Sherlock, R.R., 2007. Comparison of the effectiveness of a nitrification inhibitor, dicyandiamide, in reducing nitrous oxide emissions in four different soils under different climatic and management conditions. *Soil Use Manag.* 23, 1–9.
- Ellis, R.A., Murphy, J.G., Markovic, M.Z., VandenBoer, T.C., Makar, P.A., Brook, J., Mihele, C., 2011. The influence of gas-particle partitioning and surface-atmosphere exchange on ammonia during BAQs-Met. *Atmos. Chem. Phys.* 11, 133–145.
- Erismann, J.W., Bleeker, A., Galloway, J., Sutton, M.S., 2007. Reduced nitrogen in ecology and the environment. *Environ. Pollut.* 150, 140–149.
- Erismann, J.W., Sutton, M.A., Galloway, J., Klimont, Z., Winiwarter, W., 2008. How a century of ammonia synthesis changed the world. *Nat. Geosci.* 1, 636–639.
- Farquhar, G.D., Wetselaar, R., Firth, P.M., 1979. Ammonia volatilization from senescing leaves of maize. *Science* 203, 1257–1258.
- Galloway, J.N., Aber, J.D., Erismann, J.W., Seitzinger, S.P., Howarth, R.W., Cowling, E.B., Cosby, B.J., 2003. The nitrogen cascade. *Bioscience* 53, 341–356.
- Galloway, J.N., Townsend, A.R., Erismann, J.W., Bekunda, M., Cai, Z.C., Freney, J.R., Martinelli, L.A., Seitzinger, S.P., Sutton, M.A., 2008. Transformation of the nitrogen cycle: recent trends, questions, and potential solutions. *Science* 320, 889–892.
- Gilhespy, S.L., Webb, J., Chadwick, D.R., Misselbrook, T.H., Kay, R., Camp, V., Retter, A.L., Bason, A., 2009. Will additional straw bedding in buildings housing cattle and pigs reduce ammonia emissions? *Biosyst. Eng.* 102, 180–189.
- Hou, Y., Velthof, G.L., Lesschen, J.P., Staritsky, I.G., Oenema, O., 2017. Nutrient recovery and emissions of ammonia, nitrous oxide, and methane from animal manure in Europe: effects of manure treatment technologies. *Environ. Sci. Technol.* 51, 375–383.
- Hou, Y., Velthof, G.L., Oenema, O., 2015. Mitigation of ammonia, nitrous oxide and methane emissions from manure management chains: a meta-analysis and integrated assessment. *Global Change Biol.* 21, 1293–1312.
- Huang, S., Lv, W.S., Blaszies, S., Shi, Q.H., Pan, X.H., Zeng, Y.J., 2016. Effects of fertilizer management practices on yield-scaled ammonia emissions from croplands in China: a meta-analysis. *Field Crop. Res.* 192, 118–125.
- Jia, G., Yan, J.Y., Huang, H., Huang, L., Liu, X.L., Wang, K.N., 2009. Effects of slow-released compound acidifiers on gastrointestinal pH an immune index of avian broilers and ammonia concentration in chicken house Chinese. *Journal of Animal Nutrition* 21, 747–754 (in Chinese with english abstract).
- Khalifa, K., Zidan, A., 2001. Effect of nitrate addition on efficient use of ammonium sulfate fertilizer on corn under saline conditions. II. Field experiment. *Commun. Soil Sci. Plant Anal.* 32, 2373–2393.
- Kim, D.G., Saggarr, S., Roudier, P., 2012. The effect of nitrification inhibitors on soil ammonia emissions in nitrogen managed soils: a meta-analysis. *Nutrient Cycl. Agroecosyst.* 93, 51–64.
- Krupa, S.V., 2003. Effects of atmospheric ammonia (NH₃) on terrestrial vegetation: a review. *Environ. Pollut.* 124, 179–221.
- Li, H., Xin, H.W., Burns, R.T., Roberts, S.A., Li, S.H., Kliebenstein, J., Bregendahl, K., 2012. Reducing ammonia emissions from laying-hen houses through dietary manipulation. *J. Air Waste Manag. Assoc.* 62, 160–169.
- Loyon, L., Burton, C.H., Misselbrook, T., Webb, J., Philippe, F.X., Aguilar, M., Doreau, M., Hassouna, M., Veldkamp, T., Dourmad, J.Y., Bonmati, A., Grimm, E., Sommer, S.G., 2016. Best available technology for European livestock farms: availability, effectiveness and uptake. *J. Environ. Manag.* 166, 1–11.
- McCorry, D.F., Hobbs, P.J., 2001. Additives to reduce ammonia and odor emissions from livestock wastes: a review. *J. Environ. Qual.* 30, 345–355.
- Mendes, L.B., Pieters, J.G., Snoek, D., Ogink, N.W.M., Brusselman, E., Demeyer, P., 2017. Reduction of ammonia emissions from dairy cattle cubicle houses via improved management- or design-based strategies: a modeling approach. *Sci. Total Environ.* 574, 520–531.
- Nakhshiniev, B., Perera, C., Biddinika, M.K., Gonzales, H.B., Sumida, H., Yoshikawa, K., 2014. Reducing ammonia volatilization during composting of organic waste through addition of hydrothermally treated lignocellulose. *Int. Biodeterior. Biodegrad.* 96, 58–62.
- Nkebiwe, P.M., Weinmann, M., Bar-Tal, A., Muller, T., 2016. Fertilizer placement to improve crop nutrient acquisition and yield: a review and meta-analysis. *Field Crop. Res.* 196, 389–401.
- Oenema, O., Velthof, G.L., 1993. Ammonia volatilization from compound nitrogen-sulfur fertilizers. In: *Optimization of Plant Nutrition*, pp. 341–349.
- Pan, B.B., Lam, S.K., Mosier, A., Luo, Y.Q., Chen, D.L., 2016. Ammonia volatilization from synthetic fertilizers and its mitigation strategies: a global synthesis. *Agric. Ecosyst. Environ.* 232, 283–289.
- Paulot, F., Jacob, D.J., Pinder, R.W., Bash, J.O., Travis, K., Henze, D.K., 2014. Ammonia emissions in the United States, European Union, and China derived by high-resolution inversion of ammonium wet deposition data: interpretation with a new agricultural emissions inventory (MASAGE-NH₃). *Journal of Geophysical Research-Atmospheres* 119, 4343–4364.
- Petersen, S.O., Hojberg, O., Poulsen, M., Schwab, C., Eriksen, J., 2014. Methanogenic community changes, and emissions of methane and other gases, during storage of acidified and untreated pig slurry. *J. Appl. Microbiol.* 117, 160–172.
- Pfromm, P.H., 2017. Towards sustainable agriculture: fossil-free ammonia. *J. Renew. Sustain. Energy* 9.
- Qiao, C.L., Liu, L.L., Hu, S.J., Compton, J.E., Greaver, T.L., Li, Q.L., 2015. How inhibiting nitrification affects nitrogen cycle and reduces environmental impacts of anthropogenic nitrogen input. *Global Change Biol.* 21, 1249–1257.
- Reis, S., Howard, C., Sutton, M.A., 2015. Costs of Ammonia Abatement and the Climate Co-benefits. Springer, Dordrecht, 978-994-017-9722-1.
- Reis, S., Pinder, R.W., Zhang, M., Lijie, G., Sutton, M.A., 2009. Reactive nitrogen in atmospheric emission inventories. *Atmos. Chem. Phys.* 9, 7657–7677.
- Robertson, G.P., Bruulsema, T.W., Gehl, R.J., Kanter, D., Mauzerall, D.L., Rotz, C.A., Williams, C.O., 2013. Nitrogen-climate interactions in US agriculture. *Biogeochemistry* 114, 41–70.
- Rosenberg, M.S., Adams, D.C., Gurevitch, J., 2000. MetaWin: Statistical Software for Meta-analysis. Sunderland, MA, USA.
- Sheppard, S.C., Bittman, S., 2015. Linkage of food consumption and export to ammonia emissions in Canada and the overriding implications for mitigation. *Atmos. Environ.* 103, 43–52.
- Sommer, S.G., Schjoerring, J.K., Denmead, O.T., 2004. Ammonia emission from mineral fertilizers and fertilized crops. *Adv. Agron.* 82, 557–622.
- Stehfest, E., Bouwman, L., van Vuuren, D.P., den Elzen, M.G.J., Eickhout, B., Kabat, P., 2009. Climate benefits of changing diet. *Climatic Change* 95, 83–102.
- Stevens, C.J., Gowing, D.J.G., Wotherspoon, K.A., Alard, D., Aarrestad, P.A., Bleeker, A., Bobbink, R., Diekmann, M., Dise, N.B., Dupre, C., Dorland, E., Gaudnik, C., Rothier, S., Soons, M.B., Corcket, E., 2011. Addressing the impact of atmospheric nitrogen deposition on Western European grasslands. *Environ. Manag.* 48, 885–894.
- Stokstad, E., 2014. Air pollution ammonia pollution from farming may exact hefty health costs. *Science* 343, 238–238.
- Sutton, M.A., Oenema, O., Erismann, J.W., Leip, A., van Grinsven, H., Winiwarter, W., 2011. Too much of a good thing. *Nature* 472, 159–161.
- Sutton, M.A., Reis, S., Riddick, S.N., Dragosits, U., Nemitz, E., Theobald, M.R., Tang, Y.S., Braban, C.F., Vieno, M., Dore, A.J., Mitchell, R.F., Wanless, S., Daunt, F., Fowler, D., Blackall, T.D., Milford, C., Flechard, C.R., Loubet, B., Massad, R., Cellier, P., Personne, E., Coheur, P.F., Clarisse, L., Van Damme, M., Ngadi, Y., Clerbaux, C., Skjoth, C.A., Geels, C., Hertel, O., Kruit, R.J.W., Pinder, R.W., Bash, J.O., Walker, J.T., Simpson, D., Horvath, L., Misselbrook, T.H., Bleeker, A., Dentener, F., de Vries, W., 2013. Towards a climate-dependent paradigm of ammonia emission and deposition. *Phil. Trans. Biol. Sci.* 368.
- Ti, C.P., Luo, Y.X., Yan, X.Y., 2015. Characteristics of nitrogen balance in open-air and greenhouse vegetable cropping systems of China. *Environ. Sci. Pollut. Control Ser.* 22, 18508–18518.
- Tilman, D., Balzer, C., Hill, J., Befort, B.L., 2011. Global food demand and the sustainable intensification of agriculture. *Proc. Natl. Acad. Sci. U. S. A.* 108, 20260–20264.
- Tilman, D., Cassman, K.G., Matson, P.A., Naylor, R., Polasky, S., 2002. Agricultural sustainability and intensive production practices. *Nature* 418, 671–677.
- Trabue, S., Kerr, B., 2014. Emissions of greenhouse gases, ammonia, and hydrogen sulfide from pigs fed standard diets and diets supplemented with dried distillers grains with solubles. *J. Environ. Qual.* 43, 1176–1186.
- Van Damme, M., Clarisse, L., Heald, C.L., Hurtmans, D., Ngadi, Y., Clerbaux, C., Dolman, A.J., Erismann, J.W., Coheur, P.F., 2014. Global distributions, time series and error characterization of atmospheric ammonia (NH₃) from IASI satellite observations. *Atmos. Chem. Phys.* 14, 2905–2922.
- Varel, V.H., Nienaber, J.A., Freely, H.C., 1999. Conservation of nitrogen in cattle feedlot waste with urease inhibitors. *J. Anim. Sci.* 77, 1162–1168.
- Wang, Y., Dong, H.M., Zhu, Z.P., Gerber, P.J., Xin, H.W., Smith, P., Opio, C., Steinfeld, H., Chadwick, D., 2017. Mitigating greenhouse gas and ammonia emissions from swine manure management: a system analysis. *Environ. Sci. Technol.* 51, 4503–4511.
- Webb, J., Audsley, E., Williams, A., Pearn, K., Chatterton, J., 2014. Can UK livestock production be configured to maintain production while meeting targets to reduce emissions of greenhouse gases and ammonia? *J. Clean. Prod.* 83, 204–211.
- Webb, J., Chadwick, D., Ellis, S., 2004. Emissions of ammonia and nitrous oxide following incorporation into the soil of farmyard manures stored at different densities. *Nutrient Cycl. Agroecosyst.* 70, 67–76.
- Wu, Y.Y., Xi, X.C., Tang, X., Luo, D.M., Gu, B.J., Lam, S.K., Vitousek, P.M., Chen, D.L., 2018. Policy distortions, farm size, and the overuse of agricultural chemicals in China. *Proc. Natl. Acad. Sci. U. S. A.* 115, 7010–7015.
- Xia, L.L., Lam, S.K., Chen, D.L., Wang, J.Y., Tang, Q., Yan, X.Y., 2017a. Can knowledge-based N management produce more staple grain with lower greenhouse gas emission and reactive nitrogen pollution? A meta-analysis. *Global Change Biol.* 23, 1917–1925.
- Xia, L.L., Lam, S.K., Yan, X.Y., Chen, D.L., 2017b. How does recycling of livestock manure in agroecosystems affect crop productivity, reactive nitrogen losses, and soil carbon balance? *Environ. Sci. Technol.* 51, 7450–7457.
- Xu, P., Koloutsou-Vakakis, S., Rood, M.J., Luan, S., 2017. Projections of NH₃ emissions from manure generated by livestock production in China to 2030 under six mitigation scenarios. *Sci. Total Environ.* 607, 78–86.
- Zhang, L., Wright, L.P., Asman, W.A.H., 2010. Bi-directional air-surface exchange of atmospheric ammonia: a review of measurements and a development of a big-leaf model for applications in regional-scale air-quality models. *Journal of Geophysical Research-Atmospheres* 115.